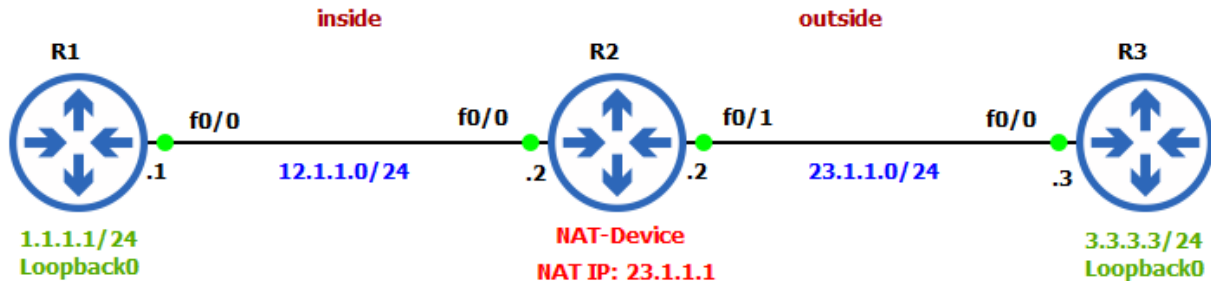


NAT Traversal:

- o NAT Traversal is a feature to enable to establish VPN session through NAT device.
- o NAT-T is a feature that allows IPsec traffic to “Traverse” through NAT or PAT points.
- o NAT-T wraps ESP packets into UDP packets and sends them across the network.



R1 Configuration

```
R1(config)#interface FastEthernet0/0
R1(config-if)#ip address 12.1.1.1 255.255.255.0
R1(config-if)#no shutdown

R1(config)#interface Loopback0
R1(config-if)#ip address 1.1.1.1 255.255.255.255
R1(config-if)#no shutdown

R1(config)#ip route 0.0.0.0 0.0.0.0 12.1.1.2

R1(config)#crypto isakmp policy 10
R1(config-isakmp)#encr 3des
R1(config-isakmp)#hash md5
R1(config-isakmp)#authentication pre-share
R1(config-isakmp)#group 2
R1(config-isakmp)#exit

R1(config)#crypto isakmp key cisco address 23.1.1.3

R1(config)#crypto ipsec transform-set TSET esp-des esp-md5-hmac
R1(cfg-crypto-trans)#exit

R1(config)#access-list 100 permit ip host 1.1.1.1 host 3.3.3.3

R1(config)#crypto map CMAP 10 ipsec-isakmp
R1(config-crypto-map)#set peer 23.1.1.3
R1(config-crypto-map)#set transform-set TSET
R1(config-crypto-map)#match address 100
R1(config-crypto-map)#exit

R1(config)#interface FastEthernet0/0
R1(config-if)#crypto map CMAP
```

R2 Configuration

```
R2(config)#interface FastEthernet0/0
R2(config-if)#ip address 12.1.1.2 255.255.255.0
R2(config-if)#no shutdown

R2(config)#interface FastEthernet0/1
R2(config-if)#ip address 23.1.1.2 255.255.255.0
R2(config-if)#no shutdown

R2(config)#ip route 1.1.1.0 255.255.255.0 12.1.1.1
R2(config)#ip route 3.3.3.0 255.255.255.0 23.1.1.3

R2(config)#interface FastEthernet0/0
R2(config-if)#ip nat inside

R2(config)#interface FastEthernet0/1
R2(config-if)#ip nat outside

R2(config)#ip nat inside source static 12.1.1.1 23.1.1.1
```

R3 Configuration

```
R3(config)#interface FastEthernet0/0
R3(config-if)#ip address 23.1.1.3 255.255.255.0
R3(config-if)#no shutdown

R3(config-if)#interface Loopback0
R3(config-if)#ip address 3.3.3.3 255.255.255.255

R3(config)#ip route 0.0.0.0 0.0.0.0 23.1.1.2

R3(config)#crypto isakmp policy 10
R3(config-isakmp)#encr 3des
R3(config-isakmp)#hash md5
R3(config-isakmp)#authentication pre-share
R3(config-isakmp)#group 2

R3(config)#crypto isakmp key cisco address 23.1.1.1

R3(config)#crypto ipsec transform-set TSET esp-des esp-md5-hmac

R3(config)#access-list 100 permit ip host 3.3.3.3 host 1.1.1.1

R3(config)#crypto map CMAP 10 ipsec-isakmp
R3(config-crypto-map)#set peer 23.1.1.1
R3(config-crypto-map)#set transform-set TSET
R3(config-crypto-map)#match address 100

R3(config)#interface FastEthernet0/0
R3(config-if)#crypto map CMAP
```

R2#show ip nat translations

Pro	Inside global	Inside local	Outside local	Outside global
udp	23.1.1.1:4500	12.1.1.1:4500	23.1.1.3:4500	23.1.1.3:4500
---	23.1.1.1	12.1.1.1	---	---

No.	Time	Source	Destination	Protocol	Length	Info
14	46.622600	23.1.1.1	23.1.1.3	ISAKMP	186	Identity Protection (Main Mode)
19	57.512365	23.1.1.1	23.1.1.3	ISAKMP	186	Identity Protection (Main Mode)
20	58.059946	23.1.1.3	23.1.1.1	ISAKMP	146	Identity Protection (Main Mode)
21	58.128292	23.1.1.1	23.1.1.3	ISAKMP	338	Identity Protection (Main Mode)
23	58.191701	23.1.1.3	23.1.1.1	ISAKMP	338	Identity Protection (Main Mode)
24	58.260303	23.1.1.1	23.1.1.3	ISAKMP	138	Identity Protection (Main Mode)
25	58.279703	23.1.1.3	23.1.1.1	ISAKMP	114	Identity Protection (Main Mode)
26	58.304315	23.1.1.1	23.1.1.3	ISAKMP	210	Quick Mode
27	58.315337	23.1.1.1	23.1.1.3	ISAKMP	210	Quick Mode

<

▶ Frame 23: 338 bytes on wire (2704 bits), 338 bytes captured (2704 bits) on interface 0

▶ Ethernet II, Src: c2:03:0f:6c:00:00 (c2:03:0f:6c:00:00), Dst: c2:02:0e:dc:00:01 (c2:02:0e:dc:00:01)

▶ Internet Protocol Version 4, Src: 23.1.1.3, Dst: 23.1.1.1

▶ User Datagram Protocol, Src Port: 500, Dst Port: 500

4 ▶ Internet Security Association and Key Management Protocol

 ▶ Payload: Vendor ID (13) : XAUTH

 ▶ Payload: NAT-Discovery (15)

 ▶ Payload: NAT-Discovery (15)

No.	Time	Source	Destination	Protocol	Length	Info
14	46.622600	23.1.1.1	23.1.1.3	ISAKMP	186	Identity Protection (Main Mode)
19	57.512365	23.1.1.1	23.1.1.3	ISAKMP	186	Identity Protection (Main Mode)
20	58.059946	23.1.1.3	23.1.1.1	ISAKMP	146	Identity Protection (Main Mode)
21	58.128292	23.1.1.1	23.1.1.3	ISAKMP	338	Identity Protection (Main Mode)
23	58.191701	23.1.1.3	23.1.1.1	ISAKMP	338	Identity Protection (Main Mode)
24	58.260303	23.1.1.1	23.1.1.3	ISAKMP	138	Identity Protection (Main Mode)
25	58.279703	23.1.1.3	23.1.1.1	ISAKMP	114	Identity Protection (Main Mode)
26	58.304315	23.1.1.1	23.1.1.3	ISAKMP	210	Quick Mode
27	58.315337	23.1.1.1	23.1.1.3	ISAKMP	210	Quick Mode

<

▶ Frame 24: 138 bytes on wire (1104 bits), 138 bytes captured (1104 bits) on interface 0

▶ Ethernet II, Src: c2:02:0e:dc:00:01 (c2:02:0e:dc:00:01), Dst: c2:03:0f:6c:00:00 (c2:03:0f:6c:00:00)

▶ Internet Protocol Version 4, Src: 23.1.1.1, Dst: 23.1.1.3

▶ User Datagram Protocol, Src Port: 4500, Dst Port: 4500

4 ▶ UDP Encapsulation of IPsec Packets

———— Last two Packets of of Phase 1 Port changed

———— New UDP encapsulation mode add

R1#show crypto isakmp sa

dst	src	state	conn-id	slot	status
23.1.1.3	12.1.1.1	QM_IDLE	1	0	ACTIVE

R3#show crypto isakmp sa

dst	src	state	conn-id	slot	status
23.1.1.3	23.1.1.1	QM_IDLE	1	0	ACTIVE